Hongsuk Benjamin Choi

PERSONAL DETAILS

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INTRODUCTION

I work on robotics and reinforcement learning. My research focuses on learning robot policies through reinforcement learning and imitation learning, including generative policies for robot control. I have been nominated for the Google PhD Fellowship starting Fall 2026.

EDUCATION & SERVICE

Ph.D., AI & Robotics — Aug 2024 - present
University of California, Berkeley
Advisor: Angjoo Kanazawa, Jitendra Malik, and Pieter Abbeel

M.S., Electrical and Computer Engineering — Mar 2020 - Feb 2022
Seoul National University, Seoul, Korea
Advisor: Kyoung Mu Lee

Military Service, Sergeant Squad Leader — Mar 2015 - Mar 2017
Air Force, Korea

B.S., Computer Science and Engineering — Mar 2015 - Feb 2020
Seoul National University, Seoul, Korea

B.S., Business — Mar 2013 - Feb 2020
Seoul National University, Seoul, Korea

PUBLICATIONS (SELECTED)

[1] Chung Min Kim, Brent Yi, David McAllister, **Hongsuk Choi**, Himanshu Gaurav Singh, Jinkun Cao, Ken Goldberg, Pieter Abbeel, Carmelo Sferrazza, Angjoo Kanazawa, "QF3: Fast Flow RL with Filtered Q-Gradients", [HOMEPAGE][PAPER]

[2] Arthur Allshire*, Himanshu Gaurav Singh*, Ritvik Singh*, Adam Rashid*, **Hongsuk Choi***, David McAllister*, Justin Yu, Yiyuan Chen, Huang Huang, Pieter Abbeel, Xi Chen, Rocky Duan, Phillip Isola, Jitendra Malik, Fred Shentu, Guanya Shi, Philipp Wu, Angjoo Kanazawa, "ABC: Scalable Behavior Cloning with Open Data, Training, and Evaluation", **CoRL 2026**, [HOMEPAGE][PAPER][CODE][DATASET] (* core contributors)

[3] Brent Yi*, **Hongsuk Choi***, Himanshu Gaurav Singh, Xiaoyu Huang, Takara E. Truong, Carmelo Sferrazza, Yi Ma, Rocky Duan, Pieter Abbeel, Guanya Shi, Karen Liu, Angjoo Kanazawa, "Flow Policy Gradients for Robot Control", **CoRL 2026**, [HOMEPAGE][ARXIV][CODE] (* equal contribution)

[4] Brent Yi, Chung Min Kim, Justin Kerr, Gina Wu, Rebecca Feng, Anthony Zhang, Jonas Kulhanek, **Hongsuk Choi**, Yi Ma, Matthew Tancik, Angjoo Kanazawa, "Viser: Imperative, Web-based 3D Visualization in Python", **RSS 2026**, [HOMEPAGE][ARXIV]

[5] David McAllister*, Songwei Ge*, Brent Yi*, Chung Min Kim, Ethan Weber, **Hongsuk Choi**, Haiwen Feng, Angjoo Kanazawa, "Flow Matching Policy Gradients", **ICLR 2026**, [HOMEPAGE][PAPER][CODE] (* equal contribution)

[6] Arthur Allshire*, **Hongsuk Choi***, Junyi Zhang*, David McAllister*, Anthony Zhang, Chung Min Kim, Trevor Darrell, Pieter Abbeel, Jitendra Malik, Angjoo Kanazawa, "Visual imitation enables contextual humanoid control", **CoRL 2025 (Oral, Best Student Paper Award)**, [HOMEPAGE][PAPER][ARXIV][CODE] (* equal contribution)

[7] Chung Min Kim*, Brent Yi*, **Hongsuk Choi**, Yi Ma, Ken Goldberg, Angjoo Kanazawa, "PyRoki: A Modular Toolkit for Robot Kinematic Optimization", **IROS 2025**, [HOMEPAGE][ARXIV][CODE] (* equal contribution)

[8] Lea Müller*, **Hongsuk Choi***, Anthony Zhang, Brent Yi, Jitendra Malik, Angjoo Kanazawa, "HSfM: Reconstructing People, Places, and Cameras", **CVPR 2025 (Highlight)**, [HOMEPAGE][ARXIV][CODE] (* equal contribution)

[9] **Hongsuk Choi***, Isaac Kasahara*, Selim Engin, Moritz Alexander Graule, Nikhil Chavan-Dafle, Volkan Isler, "FineControlNet: Fine-level Text Control for Image Generation with Spatially Aligned Text Control Injection", **WACV 2025**, [HOMEPAGE][ARXIV][CODE] (* equal contribution)

[10] **Hongsuk Choi**, Nikhil Chavan-Dafle, Jiacheng Yuan, Volkan Isler, Hyunsoo Park, "HandNeRF: Learning to Reconstruct Hand-Object Interaction Scene from a Single RGB Image", **ICRA 2024**, [HOMEPAGE][ARXIV][CODE][VIDEO]

[11] **Hongsuk Choi***, Hyeongjin Nam*, Taeryung Lee, Gyeongsik Moon, Kyoung Mu Lee, "Rethinking Self-Supervised Visual Representation Learning in Pre-training for 3D Human Pose and Shape Estimation", **ICLR 2023**, [ARXIV][VIDEO] (* equal contribution)

[12] Gyeongsik Moon, **Hongsuk Choi**, Sanghyuk Chun, Jiyoung Lee, Sangdoo Yun, "Three Recipes for Better 3D Pseudo-GTs of 3D Human Mesh Estimation in the Wild", **CVPR 2023 Workshop**, [HOMEPAGE][ARXIV][PDF]

[13] **Hongsuk Choi***, Gyeongsik Moon*, Matthieu Armando, Vincent Leroy, Kyoung Mu Lee, Gregory Rogez, "MonoNHR: Monocular Neural Human Renderer", **3DV 2022**, [ARXIV][VIDEO] (* equal contribution)

[14] **Hongsuk Choi**, Gyeongsik Moon, JoonKyu Park, Kyoung Mu Lee, "Learning to Estimate Robust 3D Human Mesh from In-the-Wild Crowded Scenes", **CVPR 2022**, [ARXIV][CODE]

[15] Gyeongsik Moon, **Hongsuk Choi**, Kyoung Mu Lee, "NeuralAnnot: Neural Annotator

for 3D Human Mesh Training Sets", **CVPR 2022 Workshop**, [ARXIV][CODE]

[16] Gyeongsik Moon, **Hongsuk Choi**, Kyoung Mu Lee, "Accurate 3D Hand Pose Estimation for Whole-Body 3D Human Mesh Estimation", **CVPR 2022 Workshop**, [ARXIV][CODE]

[17] JoonKyu Park*, Yeonguk Oh*, Gyeongsik Moon*, **Hongsuk Choi**, Kyoung Mu Lee, "HandOccNet: Occlusion-Robust 3D Hand Mesh Estimation Network", **CVPR 2022**, [ARXIV][CODE] (* equal contribution)

[18] **Hongsuk Choi**, Gyeongsik Moon, Ju Yong Chang, Kyoung Mu Lee, "Beyond Static Features for Temporally Consistent 3D Human Pose and Shape from a Video", **CVPR 2021**, [ARXIV][VIDEO][CODE]

[19] **Hongsuk Choi***, Gyeongsik Moon*, Kyoung Mu Lee, "Pose2Mesh: Graph Convolutional Network for 3D Human Pose and Mesh Recovery from a 2D Human Pose", **ECCV 2020**, [ARXIV][PDF][VIDEO][CODE] (* equal contribution)

WORK EXPERIENCE

Samsung AI & Robotics Center, New York, USA
Researcher, June 2022 - Feb 2024

Projects

- FineControlNet: Fine-level Text Control for Image Generation with Spatially Aligned Text Control Injection
- HandNeRF: Hand-Object Interaction Scene Reconstruction from a Single RGB image
- Metric-scale Human Reconstruction from a Single RGB image
- Hand Force Estimation from Vision
- User-friendly Floor Plan Generation from Lidar Image
- Manipulation Robot System Development; integrated a perception module that estimates 6D object poses from RGBD pointclouds
- Reactive Human Tracking/Following with Mobile Robots

[Tech stack]
Tekscan tactile force sensing, Vicon motion capture, Pointcloud processing using pointnet++, 6D object pose estimation from RGBD sensor, ROS2, Multi-view camera calibration and synchronization, Python 3D simulation, Feature-based NeRF, 3D hand/body reconstruction, Diffusion-based Generative AI, Mask R-CNN, etc

NAVER AI LAB, Seoul, Korea
Visiting Researcher, Mar 2022 - May 2022

- Project: Data Augmentation for 3D Human Pose and Shape Estimation using Image Generation and Pseudo-labeling
- NAVER AI LAB is the largest AI lab in Korea

[Tech stack]
Style Gan, Inpainting, Image Stitching, 3D Rendering, 3D Model Fitting, Transformer, GraphCNN, CNN, Learning-based fitting, etc

NAVER LABS Europe, Grenoble, France

Research Intern, Apr 2021 - Oct 2021

- Project: Neural Image Generation for an Arbitrary Person with Implicit Function (NeRF) from a Single RGB Image
  Advisor: Gregory Rogez, Vincent Leroy
- NAVER LABS Europe is a former Xerox Research Centre Europe

[Tech stack]
Generalizable NeRF, Sparse 3D CNN, GAN, 3D human pose and mesh estimation,3D mesh rendering/rasterization, etc

Seoul National University, Seoul, Korea
Teaching Assistant, Mar 2019 - Dec 2020

- Lecture: Introduction to Computer Programming

[Tech stack]
Python, Docker

NAVER Webtoon, Seoul, Korea
Software Engineer Intern, Jun 2018 - Aug 2018

- Project: Android App Development for Movie Trailer Play in AR
- NAVER Webtoon has 85.6 million users

[Tech stack]
Java+Android Studio, OpenGL rendering, Vuforia AR API

Start-Up
Team Leader & Software Engineer, Jul 2017 - Feb 2019

- WebApp Development for Workout Social Media
- Publicly launched at App Store & Google Play Store

[Tech stack]
Tech stack: Firebase+Node.js, TypeScript+Angular JS+Ionic Framework

PATENTS

In Process
[1] FineControlNet: FINE-LEVEL TEXT CONTROL FOR IMAGE GENERATION (U.S. Provisional Application(s), No. 63/600,563, filed in November 17, 2023 by Samsung)

[2] HandNeRF: RGB-BASED HAND-OBJECT INTERACTION RECONSTRUCTION SYSTEM (United States Patent Application No. 63/460,995, filed in April 21, 2023 by Samsung)

[3] MonoNHR: THREE DIMENSIONAL RENDERING SYSTEMS AND METHODS FROM MONOCULAR IMAGE (U.S. Provisional Application(s), No(s) 63/394,741, filed in August 3, 2022 by Seoul National University & NAVER)

PROJECTS

Human mesh model parameter fitting from 3D point cloud — Mar 2021 - Oct 2021
co-op with Antigravity

A Large-scale Dataset for 3D Human Pose and Mesh Estimation May 2020 - Dec 2020
Dataset link / co-op with SweetK and MotionTechnology

HumanFit: A New Large-scale Dataset for Human Fitness Evaluation and Feedback May 2020 - Dec 2020
Dataset link / co-op with SuperbAI, KakaoBrain

Human tracking and counting from Drone images (Data collection and Model development) Mar 2019 - May 2019
co-op with Ministry of Science and ICT, Korea

Human Part Segmentation for AI-based Animation Coloring Mar 2019 - Jun 2019
co-op with NCSOFT

HONORS

Best Student Paper Award at CoRL 2025 for "Visual imitation enables contextual humanoid control", 2025

Distinguished Master Dissertation Award, Seoul National University, 2022

Selected as a finalist in **Qualcomm Innovation Fellowship Korea**, 2020 & 2021

1st place and **2nd place** at the *without association* track of **3D human pose estimation in the wild (3DPW) challenge**, workshop conjunction with **ECCV 2020** (1st in a joint orientation metric and 2nd in a joint position metric), 2020

1st place at the Qualcomm IT Tour presentation competition held by Qualcomm, proposed AR-based workout coaching system and selected as a winner by Jim Cathey (President of Qualcomm Global Business), 2019

SKILLS

Professional at:
Python, Pytorch, Pytorch3D, OpenCV, Open3D, PIL,Camera Calibration, Human Body and Hand Motion Capture, Vicon, 3D Visualization, Performance Report, Conventional Multi-View Geometry CV, Image Processing, Machine Learning, Convex Optimization, Latex

Familiar at:
Java, Javascript, Tensorflow, ROS, Android Studio, Python Simulation, Tactile Force Scanning, 3D Graphics

SOMETHING DIFFERENT

During my stay in Bristol, UK, I went by Benjamin in an elementary school. I played an electric guitar in a university rock band. And long after that I played a bass guitar and jammed with friends in New York.